MATH3018 Assignment 8, due on Apr 28

(1) Consider two sample sets X_1 and X_2 from two bivariate normal populations C_1 and C_2, respectively:

$$X_1 = \begin{pmatrix} 3 & 7 \\ 2 & 4 \\ 4 & 7 \end{pmatrix}, X_2 = \begin{pmatrix} 6 & 9 \\ 5 & 7 \\ 4 & 8 \end{pmatrix}.$$

Assume a same population covariance matrix for C_1 and C_2.
(a) Calculate the linear classification function $\delta_{12}(x)$.
(b) Classify the observation $x_0 = (2, 7)^T$.

(2) Let $f_1(x) = P(X = x|Y = 1) = \frac{1}{2}(1 - |x|)$ for $|x| \leq 1$ and $f_2(x) = P(X = x|Y = 2) = \frac{1}{2}(1 - |x - .5|)$ for $-.5 \leq x \leq 1.5$.
(a) Sketch the two densities.
(b) Identify the classification regions of Bayes classifier when the prior probability of class $\pi_1 = \pi_2$.
(c) Identify the classification regions when $\pi_1 = .2$

(3) Consider random samples from three bivariate normal populations C_1, C_2, and C_3 with the same population covariance matrices. The sample mean vectors and covariance matrices, are as follows:

$$C_1 : n_1 = 3, \quad \bar{\mathbf{x}}_1 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}, \quad \mathbf{S}_1 = \begin{bmatrix} 1 & -1 \\ -1 & 4 \end{bmatrix}$$

$$C_2 : n_2 = 3, \quad \bar{\mathbf{x}}_2 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \quad \mathbf{S}_2 = \begin{bmatrix} 1 & -1 \\ -1 & 4 \end{bmatrix}$$

$$C_3 : n_3 = 3, \quad \bar{\mathbf{x}}_3 = \begin{bmatrix} 0 \\ -2 \end{bmatrix}, \quad \mathbf{S}_3 = \begin{bmatrix} 1 & 1 \\ 1 & 4 \end{bmatrix}$$

Given that $\pi_1 = \pi_2 = 0.25$, $\pi_3 = 0.5$, classify the observation $x_0 = (2, -1)^\top$ using QDA.

(4) Suppose that we have prior probability for two classes C_1 and C_2 given as π_1 and π_2, and the density of $\mathbf{x}$ in C_1 and C_2 are given as $f_1(x)$ and $f_2(x)$, respectively. Show that by classifying those $\mathbf{x}$ satisfying $\pi_1 f_1(\mathbf{x}) \geq \pi_2 f_2(\mathbf{x})$ as C_1 and $\pi_1 f_1(\mathbf{x}) < \pi_2 f_2(\mathbf{x})$ as C_2 achieves the minimum value of TPM (total probability of misclassification).